

Days 1-5

Build and Configure a Small Office Network Deliverable : Fully functional LAN with: Proper IP addressing and subnetting. Configured DHCP and DNS services. Lans for network segmentation. Simulated internet access via a router. Troubleshooting documentation for common issues. Task : Submit a detailed report on the network setup and configurations performed.

1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

ANS :

Network Details:

- Network: 192.168.1.0/24
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Device IP Configuration:

Device	IP Address	Subnet Mask	Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.11	255.255.255.0	192.168.1.1
PC3	192.168.1.12	255.255.255.0	192.168.1.1
PC4	192.168.1.13	255.255.255.0	192.168.1.1
PC5	192.168.1.14	255.255.255.0	192.168.1.1

Verification:

- Used ping command between all PCs.
- All devices successfully replied with **0% packet loss**.
- Confirmed successful LAN communication.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

ANS :

DHCP Server Settings:

- DHCP Pool Name: OFFICE_POOL
- Network: 192.168.1.0
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Start IP: 192.168.1.20
- Max Users: 10

Assigned IPs (Automatic):

Device	IP Address
Laptop1	192.168.1.20
Laptop2	192.168.1.21
Laptop3	192.168.1.22

Verification:

- Devices successfully received IPs via DHCP.
- Checked using ipconfig command.
- All devices communicated within LAN successfully.

3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface. Hint : You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

ANS :

DNS Setup:

- DNS Server IP: 192.168.1.2
- Domain Mapping:
musicplayer.local → 192.168.1.30

Test Device:

- Web Server IP: 192.168.1.30

Result:

- Entered http://musicplayer.local in browser.
- Successfully opened the local web page hosted on server.
- DNS resolution working correctly.

4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN. Hint : Use ping or file sharing to test connectivity between devices in different VLANs.

ANS :

VLAN Setup:

VLAN ID	Name	Devices
VLAN 10	Workstations	PC1, PC2
VLAN 20	Smart Devices	PC3, PC4

Switch Configuration (Concept):

- VLAN 10 created and assigned to ports 1-2
- VLAN 20 created and assigned to ports 3-4

Testing Results:

Test	Result
PC1 ↔ PC2 (same VLAN)	Successful
PC3 ↔ PC4 (same VLAN)	Successful
VLAN 10 ↔ VLAN 20 communication	Failed (Expected)

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

ANS :

Router Configuration:

- Inside Network: 192.168.1.0/24
- Outside Interface: Connected to ISP network
- NAT Enabled: Yes (Overload/PAT used)

Testing:

- Ping to external website: www.wikipedia.org
- Result: Successful response received
- Web browsing simulation: Working

Conclusion:

- NAT correctly translates private IPs to public IP.
- LAN devices can access external network successfully.